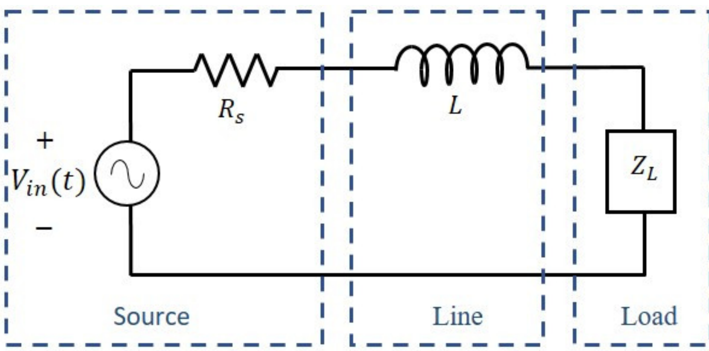
Lab 10 - Power Transfer in AC Circuits

Objective
The goal of this lab is to familiarize students with the process of delivering power to a load. We will focus, in particular, on the problem of how to choose the load impedance to maximize the amount of power to the load and the impact if the load impedance is not chosen optimally.

Introduction
In this lab, you will investigate the issue of transferring power from an AC voltage source to a load. The situation is illustrated below where the source is modeled as an AC voltage source with series resistance R_s. Typically the source is connected to the load through some line which may have a certain amount of line inductance which we can model as an inductor between the source and the load. Often, the load is resistive and in which case we will model it as a load resistor, R_L. As we discussed in lecture, adding a "shunt" capacitor can help improve the power transfer, which is also shown below.

Case 1: Resistive Load Only



To start with, consider the case where the load is purely resistive so that Z_L=R_L. Since the source is sinusoidal, we can use phasors to calculate the load voltage through a voltage divider as:

$$V_L = V_{\text{in}} \frac{R_L}{(R_S + R_L) + j(\omega L)}$$

Now, the average power delivered to the load can be determined :

$$P_L = \frac{1}{T_o} \int_{T_o} \frac{v_L^2(t)}{R_L} dt = \frac{V_m}{2 R_L}$$

using the identity that the mean square value of a sinusoid is 1/2 the max. Combining this with the previous equation yields:

$$P_L = \frac{V_L^2}{2 R_L} = \left(\frac{V_{\text{in}}^2}{2 R_L}\right)\left(\frac{R_L^2}{(R_s + R_L)^2 + (\omega L)^2}\right) = \left(\frac{V_{\text{in}}^2}{2}\right)\left(\frac{R_L}{[(R_s + R_L)^2 + (\omega L)^2]}\right)$$

Note that all of these values are a constant except for R_L. Thus, to find the max power transfer, we can set dP_L/dR_L = 0 and solve for R_L, thus:

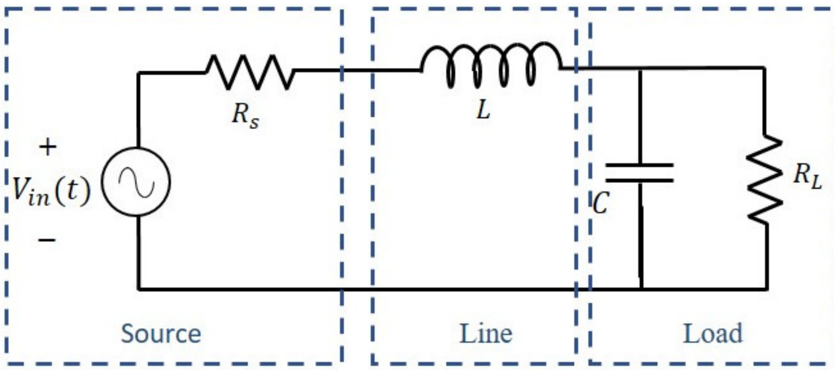
$$\frac{d}{dR_L}P_L=\left(\frac{V_{\text{in}}^2}{2}\right)\left(\frac{-R_L^2+R_s^2+(\omega L)^2}{[(R_s+R_L)^2+(\omega L)^2]^2}\right)=0\Rightarrow R_L=\sqrt{R_s^2+(\omega L)^2}$$

Solving for V_{in}=5V, R_s=100Ω, L=0.033H, and f=5000Hz yields:

$$R_L=\sqrt{100^2+(31.4k*0.033)^2}=1041.5\Omega$$
$$\omega L=(2\pi)(5000)(0.033)=1036.7\,\Omega$$
$$P_L=\frac{5^2}{2}\frac{1041.5}{(1041.5+100)^2+(1036.7)^2}=0.005458\,W=5.5\,mW$$

Note, P_L is the average power delivered as the actual power delivered is sinusoidal.

Case 2: Resistive Load with Shunt Capacitor



As quite often is the case, the load resistance may be fixed and not under our control. For example, if the load is our desk lamp or maybe a toaster, there is not much we can do to change its resistance. In such a case, it is sometimes beneficial to place what is known as a shunt capacitor between the terminals of the load. In this case, we treat the load resistance as fixed and seek the value of the shunt capacitance that will cause the maximum power to be delivered to the load resistance.

Once again applying a voltage divider formula (and a bit of algebra), we can determine that the load voltage of the circuit is:

$$V_L=V_{\text{in}}\frac{R_L}{(R_S+R_L-\omega^2 R_L LC)+j\,\omega(L+R_L R_S C)}$$

and the average power is:

$$P_L=\frac{V_L^2}{2R_L}=\left(\frac{V_{\text{in}}^2}{2}\right)\left(\frac{R_L}{(R_S+R_L-\omega^2 R_L LC)^2+[\omega(L+R_L R_S C)]^2}\right)$$

In this case, all of these values are again a constant except for C. Thus, to find the max power transfer, we can set dP_L/dC = 0 and solve for C. After much algebra. we find the shunt capacitor value for maximum power transfer to be:

$$C=\frac{L}{\omega^2 L^2+R_s^2}$$

Interestingly, C only depends on the source components (ω, L, and R_s) and not the load!

Once again, solving for V_{in}=5V, R_s=100Ω, L=0.033H, and f=5000Hz, and setting R_L = 1041.5Ω (from Case 1) yields:

$$C=\frac{0.033}{[(2\pi)(5000)]^2[.033]^2+(100)^2}=30.42nF$$

and the power transfer to the load is:

$$\text{let } D_1=R_S+R_L-\omega^2 R_L LC=100+1041.5-4\pi^2(5000)^2(1041.5)(0.033)(30.42\,x\,10^{-9})=109.6\,\Omega$$

$$\text{let } D_2=\omega(L+R_L R_S C)=2\pi(5000)(0.033+(1041.5)(100)(30.42\,x\,10^{-9}))=1136.3\,\Omega$$

$$P_L=\left(\frac{V_{\text{in}}^2}{2}\right)\left(\frac{R_L}{D_1^2+D_2^2}\right)=\frac{5^2}{2}\frac{1041.5}{(10.96)^2+(1136.3)^2}=0.00999\,W=9.99\,mW$$

Thus, by adding a 30nF shunt capacitor to our load (irrespective of the load value!), we were able to deliver twice as much average power to the load then without the shunt.

Procedure

Parts and equipment needed:

- 100Ω resistor and 33mH inductor
- 7-8 various resistors between 10Ω and 10kΩ (must include 1kΩ)
- 5-6 various capacitors between (must include 47nF and 100nF)
- A selection of colored 22 gauge connection wires
- XK700 Trainer
- Oscilloscope

Case 1: Resistive Load Only

Build the circuit from the theory section setting V_{in}=5V@5kHz, R_s=100Ω and L=33mH. Use the selection of resistors you chose as R_L. For each R_L, record its resistance and measured V_L.

Case 2: Fixed Resistor with Variable Capacitor

- Build the circuit from the theory section setting V_{in}=5V@5kHz, R_s=100Ω, L=33mH, and R_L=1kΩ. Use the selection of capacitors you chose as C. For each C, record its capacitance and measured V_L. To get close to the optimal shunt capacitance, put the 47nF and 100nF in series (and parallel with R_L).
- Repeat the above using 2 other resistors from Case 1 as your R_L=1kΩ (one above and one below 1kΩ).

Lab Report

- Title page
- Procedure: Write in your own words what you did in lab
- Data and Results:
  - Calculate the power transferred for each circuit.
  - Provide a table of R_L, V_L, and P_L for each circuit in case 1. Provide a plot P_L vs R_L.
  - Provide a table of C, V_L, and P_L for each circuit in case 2. Provide a plot P_L vs C for a 3 R_L values (same plot).
- Discussion:
  - What kind of deviation are you seeing between the measured and theoretical results? If there are differences that are significant, be sure to discuss the most likely sources of these errors.
  - Comment on how the dissipated power improves as we optimize the various component values. How much difference does it make? How did the power change?